

Predicting Offense Levels Across NYPD Arrests, 2023-25

Research by: Areebah Aziz, Ashley Brea, Luke Jenkins

Link to Github Rehub:

<https://github.com/ashleybrea/Weighted-Scales-Demographic-Disparities-in-NYPD-Arrests>

Introduction & Motivation

Systemic racism and socioeconomic inequality shapes who gets arrested, for what charges, and how seriously those arrests are charged within the NYPD's criminal system. Arrests made by NYPD officers have received criticism and made headlines for excessive force as well as potential demographic biases towards marginalized groups. Understanding whether and how demographic factors correlate with offense level at the point of arrest is essential for evaluating whether NYPD enforcement practices reflect systemic disparities or borough-specific policing practices. Data on crime rates and arrests conducted by the NYPD are helpful in determining the work that needs to be done to generally reduce crimes in NYC, as well as if current policies towards crimes are helpful in solving such a vast problem. Using NYPD Historic Arrest Data (2006-Present), the question we are answering is: How does the relationship between demographic information and level of offense (felony, misdemeanor, violation) vary across NYC's boroughs?

Currently, research has been done on NYPD arrests using predictive analytics to identify patterns and to forecast future crime occurrences. Previous research has found that drug-related offenses committed by people aged 25-44 are the most common in NYC ([Kumar et al. 2024](#)). While Kumar et al.'s research focuses on general criminal trends in hotspots and contextualization based on socioeconomic status, we aim to conduct this research with the goal of uncovering biases. Furthermore, while this research uses NYPD data from 2006-2017, we will be contextualizing criminal offenses from 2023-2024, which can provide insight on recent social trends and crime rates. Similarly, racial biases within NYPD arrests have been researched, specifically the use of the Stop-and-Frisk program and creating a bias-free dataset to prevent future biased stops ([Badr & Sharma, 2022](#)). Our research project will use similar techniques to both of these research papers, utilizing K-Nearest Neighbors and Random Forest Classification, as well as the construction of neural nets using scikit-learn's Multi-layer Perceptron classifier (MLP).

By conducting this research, we aim to uncover biases within NYPD arrests and contribute our analysis to research that can predict future trends involving crime rates and offense levels, and better understand arrests classification's impact on marginalized groups. Our goal with this project is to analyze the relationship between demographic information, like race and gender, and level of criminal offense. We hypothesize that marginalized groups, particularly Black and Hispanic individuals, are disproportionately arrested for lower-level offenses relative to other racial groups, and that this disparity will be consistent across NYC boroughs and precincts, reflecting a systemic rather than localized pattern. We expect that race, age, and sex will emerge as significant predictors of offense level in both our models, and that comparing these models' performance will reveal which approach better captures these demographic relationships. Ultimately, we aim to provide data-driven evidence that either confirms or challenges the

presence of systemic demographic disparities in how NYPD arrests are classified.

Data Description

Our primary [dataset](#), entitled “NYPD Arrests Data (Historic)” from [NYC OpenData](#), includes all NYPD arrests made to date since 2006. The data was provided by the NYPD itself and contained nearly 6 million entries before cleaning.

Initial Cleaning

Given the sheer size of the original dataset, we began by focusing on a singular ten year span (2015 to 2025), so we could potentially explore specific questions related to time periods. We queried our data when extracting it from the website to only include arrests from this ten year span. We removed rows with missing values in any of our above columns, and rows that described the arrests of minors. Our ARREST_DATE was transformed to a datetime data type, and our other variables were cast to a category data type. To help us easily focus on certain parameters, we removed columns we do not foresee using in future analyses, including various classification codes the NYPD included and multiple ways to measure latitude and longitude where a crime took place (although these columns might be of interest to researchers in the future).

Further Data Preparation

During the modeling process, we realized additional cleaning was required beyond our initial proposal phase. Upon inspecting LAW_CAT_CD further, we found that in addition to the expected classes (F, M, V), the column contained invalid values such as “I”, “9”, and “null”. These were filtered out before training. We also confirmed that LAW_CAT_CD was the correct target variable for our research questions rather than LAW_CODE, which has thousands of varying law codes, which we thought was not appropriate for our classification.

After realizing the categorical nature of some of features would be incompatible with the models we were attempting to train, we decided to convert those features (PERP_RACE, PERP_SEX, AGE_GROUP, and ARREST_BORO) using the pandas method `.get_dummies()`, understanding that this comes with potential limitations of creating a sparse dataframe that is slower to train. We originally did the dummy creation manually, but then recalled this method that makes processing more efficient. These variables take values of either 0 or 1, and are labeled in the same fashion: with “IS” followed by an underscore and the associated demographic characteristic. They are used in all models and central to our exploration to see if they are valuable predictors for the offense level of an arrest. For testing across sex, we have one dummy variable “IS_MALE”, so our analysis in this regard is limited to those arrested who were male, and those arrested who were not male.

A key decision in our data preparation was narrowing our “ARREST_DATE” date range to 2023–2024, rather than using the full 2015–2025 span initially extracted. We made this choice primarily for computational reasons since training our models on a decade of data proved too time-intensive on our personal computers utilizing Google Colab. We prioritized training on the two most recent complete years

of data. This also has an analytical advantage since 2023–2024 reflects the current policing environment, making our results more applicable to evaluating contemporary NYPD practices. Due to the sheer size of the dataset and the number of calculations needed to train certain models, like K- Fold Cross Validation, we narrowed the data set before this part of KNN training. Besides slimming down our focus to the years 2023-2024, we used random sampling to narrow the number of data points used in training to 100,000, and tested values of k from 1 to 20.

Variable descriptions (when sourced from NYC OpenData):

ARREST_DATE	Exact date of arrest for the reported event
LAW_CAT_CD	Level of offense: felony(F), misdemeanor(M), violation(V)
ARREST_BORO	Borough of arrest. B(Bronx), S(Staten Island), K(Brooklyn), M(Manhattan), Q(Queens)
ARREST_PRECINCT	Precinct where the arrest occurred
AGE_GROUP	Perpetrator's age within a category
PERP_SEX	Perpetrator's sex description
PERP_RACE	Perpetrator's race description

Methods

We decided to pursue four different model types in our data analysis. We were introduced to Random Forest Classifiers and its associated adaptability through our scholarly research, and knew we could also successfully implement Logistic Regression, Neural Nets, and KNN to perform our multiclass classification with discrete numeric variables. We assessed each model's performance not only overall but broken down by demographic subgroups — race, sex, and borough. A model that may appear accurate overall may still systematically misclassify arrests within specific demographic subgroups, which is precisely the kind of disparate impact we set out to investigate. If accuracy varied significantly across

racial or geographic subgroups, that would suggest the model was capturing real demographic signals.

Random Forest Classifier

Our research required going beyond the learned models in our course, which involved the use of a Random Forest Classifier—a model that constructs a large number of decision trees (a “forest”) during training and combines their results to reach a more accurate prediction, usually by taking a majority vote. Each tree is built on a random subset of features and a bootstrapped sample of the data, which reduces overfitting. We selected this method because it handles non-linear relationships between features, provides feature importance rankings, and is well-suited to multiclass classification.

Model Implementation

We initially trained a baseline Random Forest Classifier (RFC) with default parameters. This resulted in an accuracy of 0.2559 but a balanced accuracy of 0.4999. This suggested to us that we should observe the underlying spread of offense levels, and learned that misdemeanors are around 57% of arrests. Violations made up less than 1%. This suggested that standard models would be heavily biased toward predicting the majority class of the data. To address the issue of class imbalance in our RFC, we retrained the model using the “class_weight=’balanced’” parameter. We then attempted some hyperparameter tuning, such as increasing n_estimators to 200, setting max_depth to 10, and setting min_samples_leaf to 5. This ended up producing similar accuracy and balanced accuracies.

KNN

Model Implementation

We trained a baseline KNN model using age, race and borough to predict offense level—initially setting our k to 1, 3, and 10. We observed that a k of 1 resulted in the best accuracy. We then decided to perform K- Fold Cross Validation using a random selection of data points to reduce calculation time without sacrificing accuracy. This resulted in a best k of 19 and a best CV error rate of 0.4639. After finding an ideal k value of 19, we observed an accuracy score of 0.5442975 over the test set. While we were not able to illustrate feature performance, we created a confusion matrix to again visualize model performance.

Logistic Regression

Model Implementation

The scikit-learn Logistic Regression algorithm was implemented with max_iter= 5000. The model gained a base accuracy score of 0.574 and a balanced accuracy of 0.254. Features were scaled to account for low accuracy, but the accuracy score didn’t change significantly. We then decided to test accuracy rates for the model across race, borough, and sex, which we also executed for the following method (neural networks).

Neural Networks Using MLP

Model Implementation

We trained a model using the MLPClassifier method using Adam as an optimization algorithm, with hidden layer sizes of 62 and 34, with max_iter=200. We also produced a confusion matrix, and calculated

accuracy by borough, race, and sex. We then attempted using SMOTE (Synthetic Minority Over-sampling Technique) to see if this would remedy any issues with the class imbalance in our data set. The accuracy did not significantly change when training a MLP with a SMOTE dataset.

Results

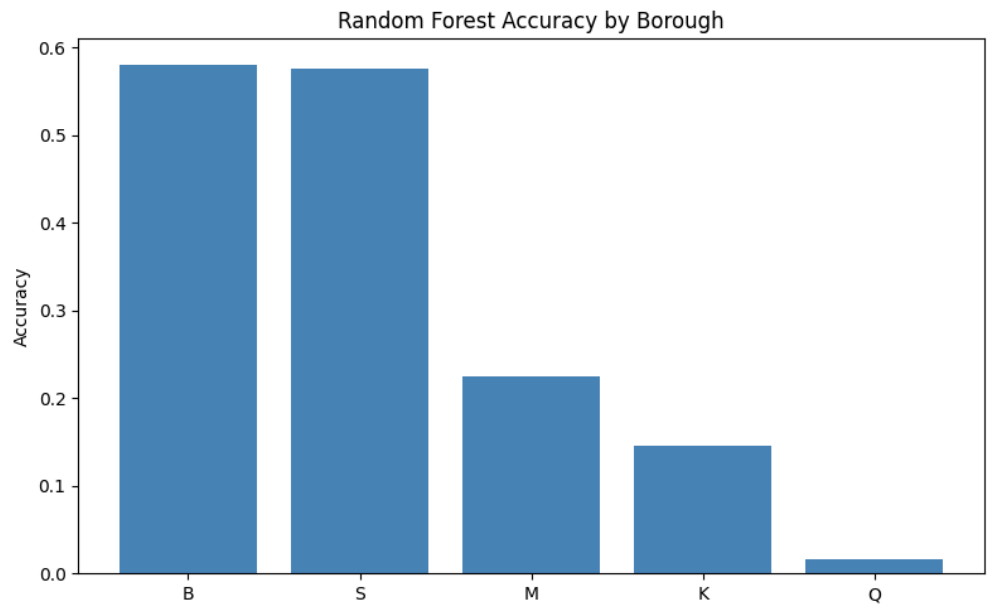
Random Forest Classifier Visualizations & Results

Visualizations/Summary Statistics

Balance Type	Accuracy	Balanced Accuracy
No Tuning	0.5752242778040091	0.2580762168445947
Tuned	0.25471090698373183	0.5038508997789022

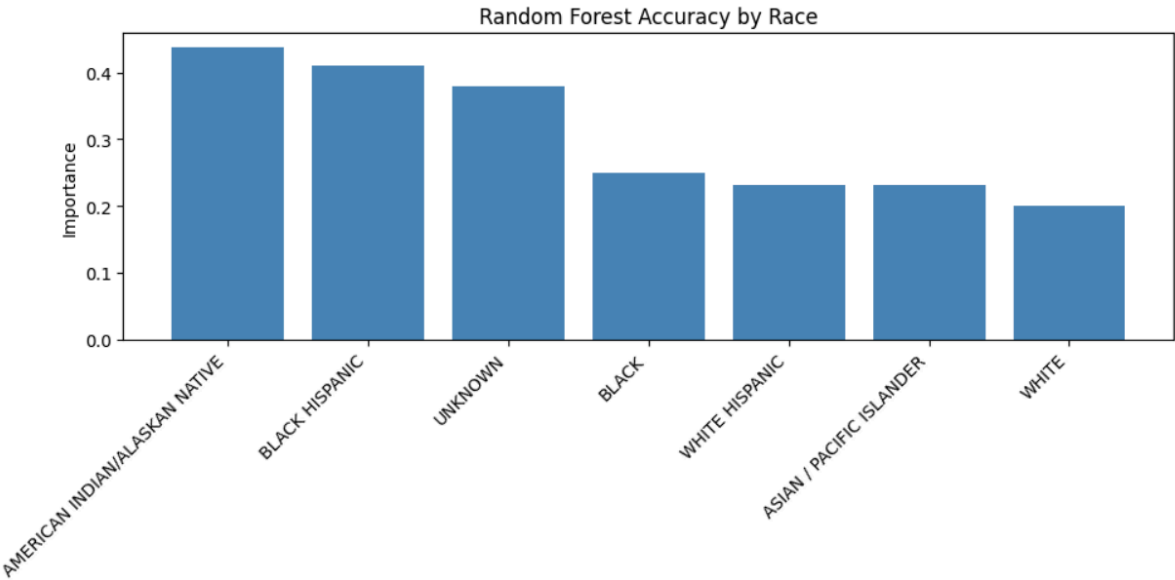
Accuracy Separated By Borough:

Borough	Accuracy	Balanced Accuracy
Queens	0.02	0.251
Bronx	0.58	0.26
Manhattan	0.225	0.307
Brooklyn	0.148	0.298
Staten Island	0.573	0.267



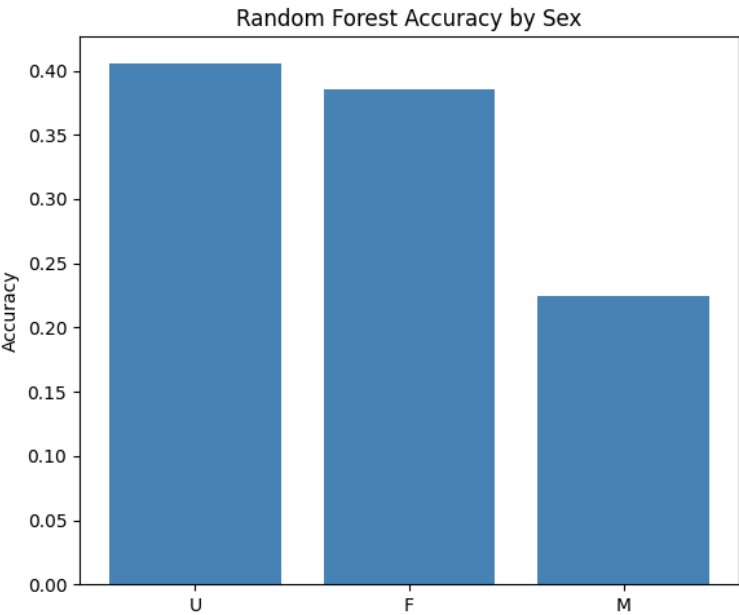
Accuracy Separated By Race:

Race	Accuracy	Balanced Accuracy
Black & Hispanic	0.411	0.502
White	0.20	0.515
AAPI	0.231	0.462
American Indian or Alaskan Native	0.437	0.562
Race unknown	0.380	0.552
Black	0.249	0.490
White & Hispanic	0.231	0.498



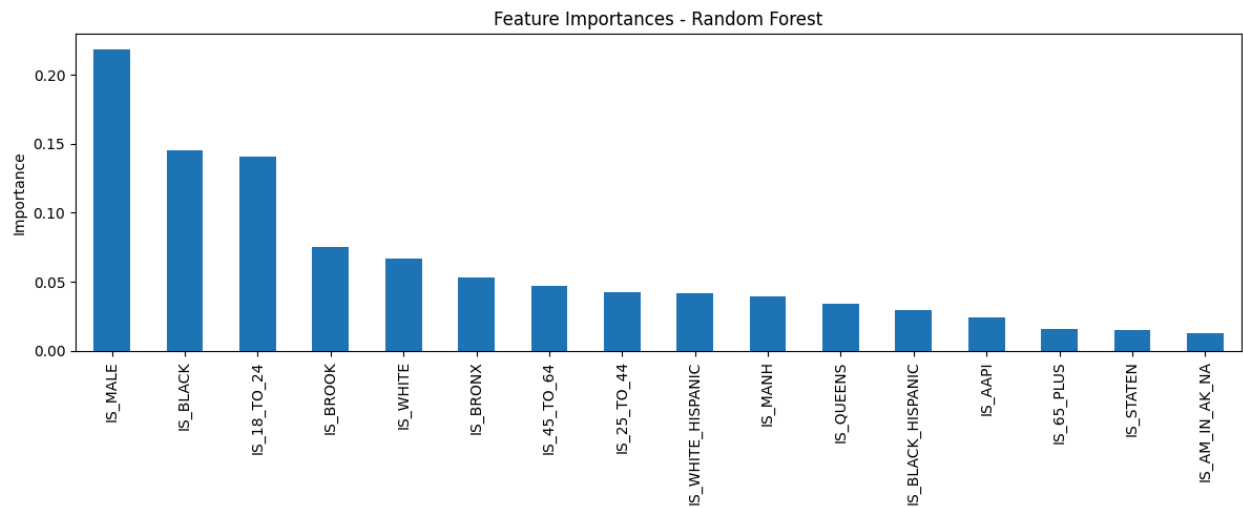
Accuracy Separated By Sex:

Sex	Accuracy	Balanced Accuracy
Male	0.22	0.49
Female	0.385	0.545
Unknown	0.406	0.218

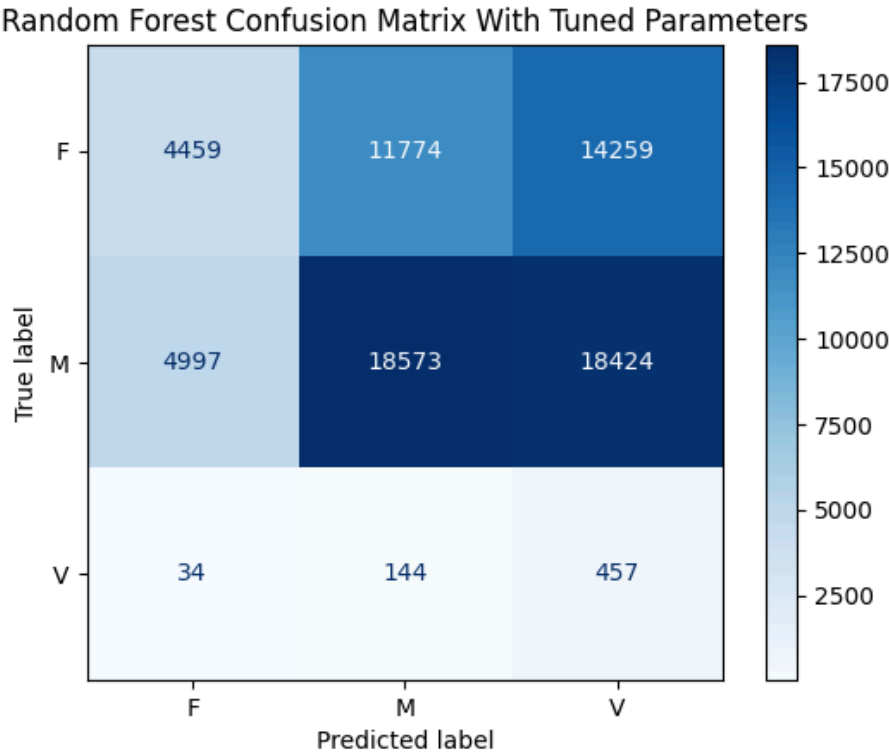


Accuracy Rates for each sex using Random Forest Classifier.

Below we visualize how important each feature was in our RFC, and our confusion matrix for the tuned RFC model. Notice the discrepancy between volumes of each class.



This bar chart shows how important each feature is and how much each feature contributes to the model’s prediction accuracy. A person who is Male, Black, and 18-24 years old are the most influential features in this model.



This confusion matrix shows the true positives, false positives, true negatives, and false negatives between the true labels and the predicted labels of an observation. Higher values show a strong performance in classifying data for a specific label.

Misdemeanors overtake the model's predictions.

KNN Visualizations & Results

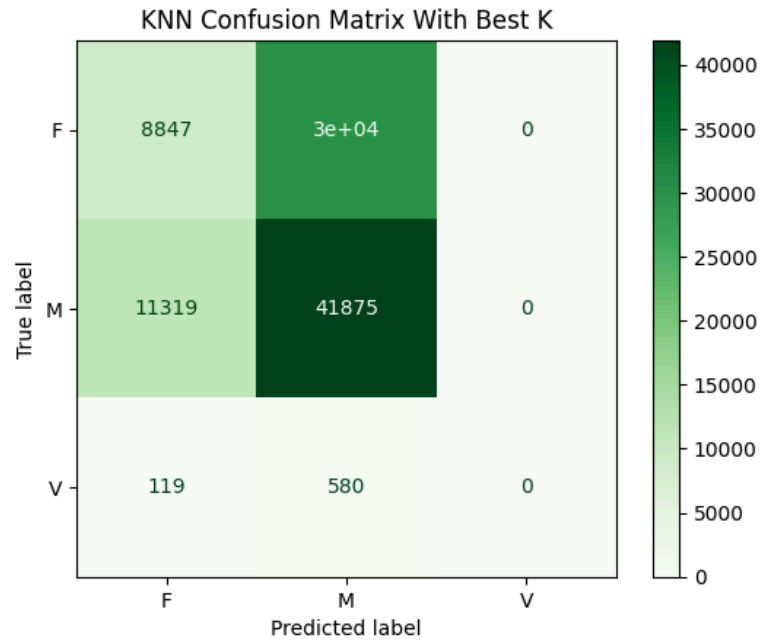
Visualizations/Summary Statistics

k value	Accuracy
1	0.49388
3	0.514
10	0.518
best k: 19	0.544

Accuracy by Offense Group

Level of Offense	Accuracy
Misdemeanor (M)	0.787213
Felony (F)	0.226730
Violation (V)	0.000000

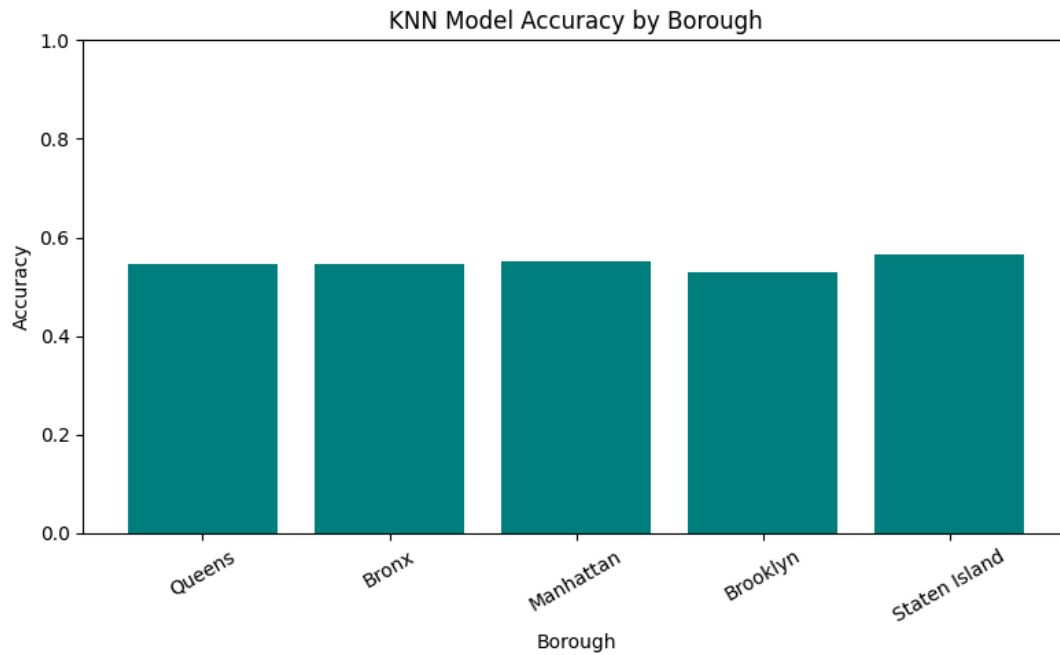
KNN Confusion Matrix with K=19



Higher values show a strong performance in classifying data for a specific label. Misdemeanors overtake the model's predictions.

Accuracy by Borough

Borough	Accuracy
Queens	0.54531
Bronx	0.54688
Manhattan	0.55279
Brooklyn	0.53067
Staten Island	0.56594

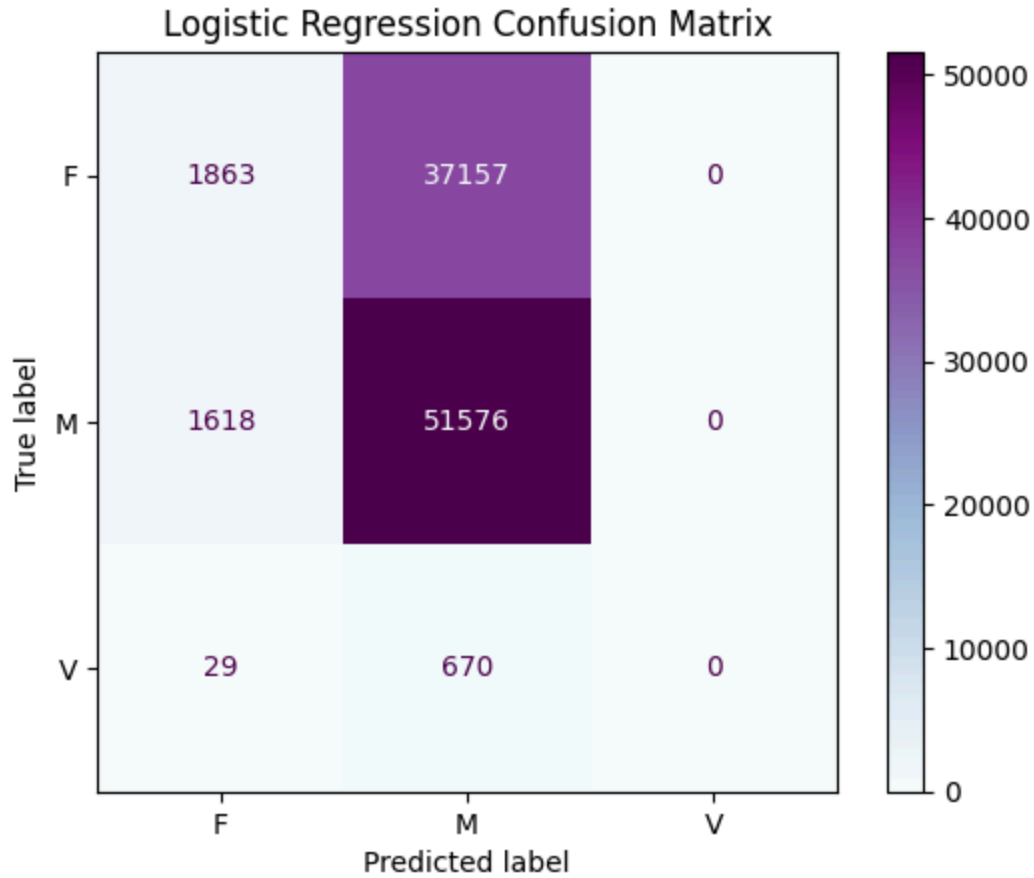


KNN Model accuracies by borough are relatively the same, in the 50-56% range.

Logistic Regression Visualizations & Results

Visualizations/Summary Statistics

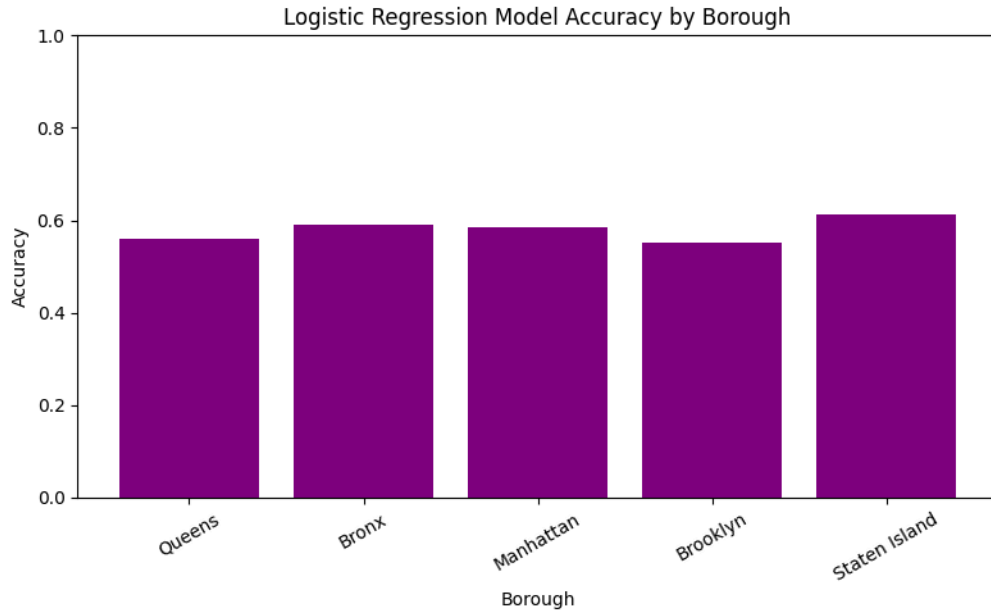
	Accuracy	Balanced Accuracy
Entire dataset	0.573453663561832	0.2543319454911682



This confusion matrix shows the true positives, false positives, true negatives, and false negatives between the true labels and the predicted labels of an observation. Higher values show a strong performance in classifying data for a specific label. Misdemeanors overtake the model's predictions.

Accuracy Separated By Borough:

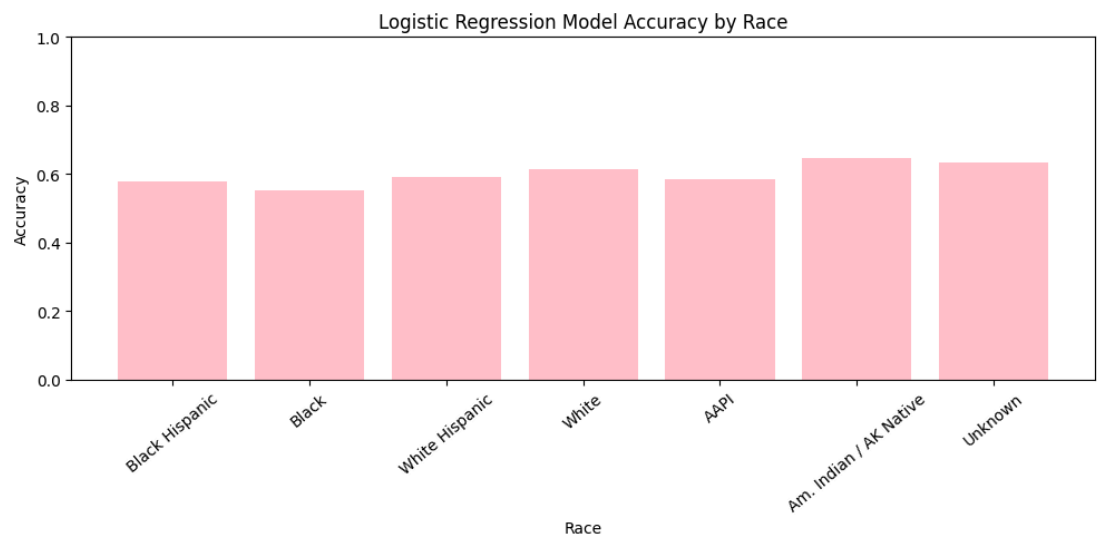
Borough	Accuracy	Balanced Accuracy
Queens	0.5613573131955485	0.2548511870910644
Bronx	0.5915301187849195	0.25
Manhattan	0.585708441292786	0.25
Brooklyn	0.5512157167866174	0.258813872862102
Staten Island	0.6130145427655903	0.25



The bar chart shows the accuracies between different boroughs for the Logistic Regression Model. Staten Island has the highest accuracy, followed by the Bronx and Manhattan.

Accuracy Separated By Race:

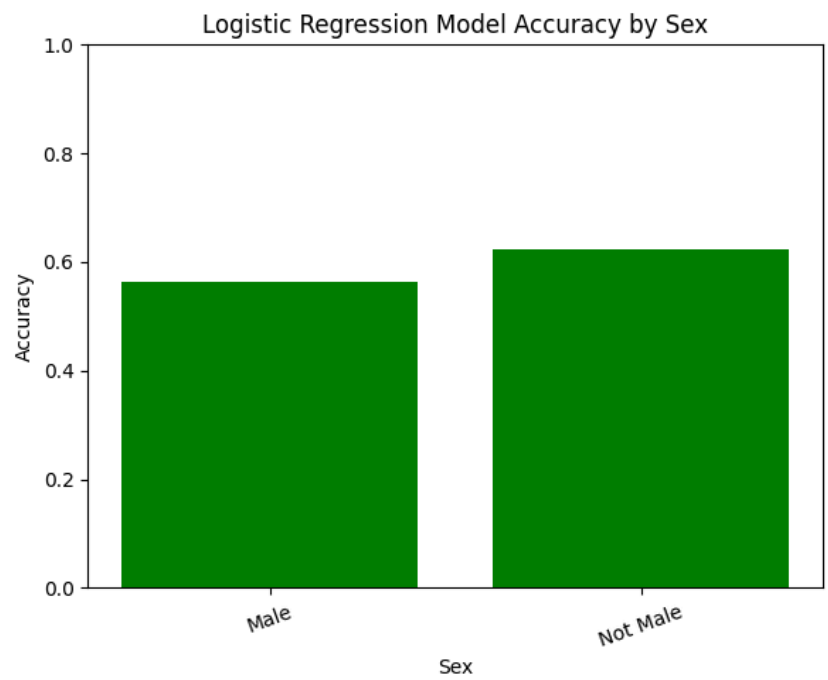
Race	Accuracy	Balanced Accuracy
Black & Hispanic	0.5772683032334811	0.25259662580997544
White	0.6123488224061108	0.25
AAPI	0.5845771144278606	0.25
American Indian or Alaskan Native	0.6472491909385113	0.25
Race unknown	0.6350148367952523	0.25
Black	0.5512870292122514	0.25666600234208037
White & Hispanic	0.5916486563634139	0.25



Bar chart showing the accuracies between different races for the Logistic Regression Model

Accuracy Separated By Sex:

Sex	Accuracy	Balanced Accuracy
Male	0.5623740997722275	0.25474252153420845
Not Male	0.6222802436901653	0.25

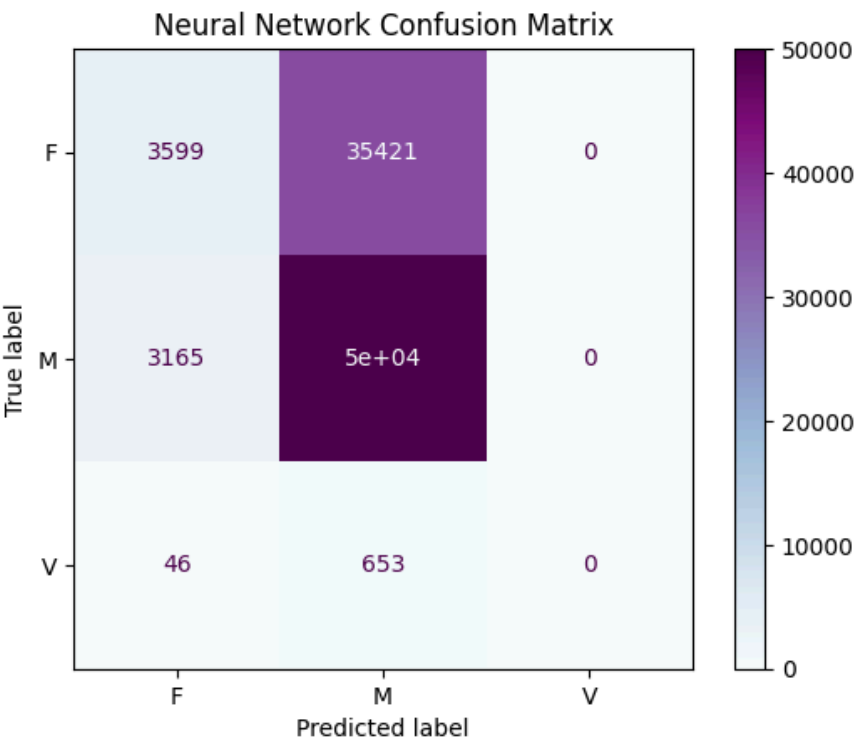


Accuracies of each sex (Male or Female) in the Logistic Regression model

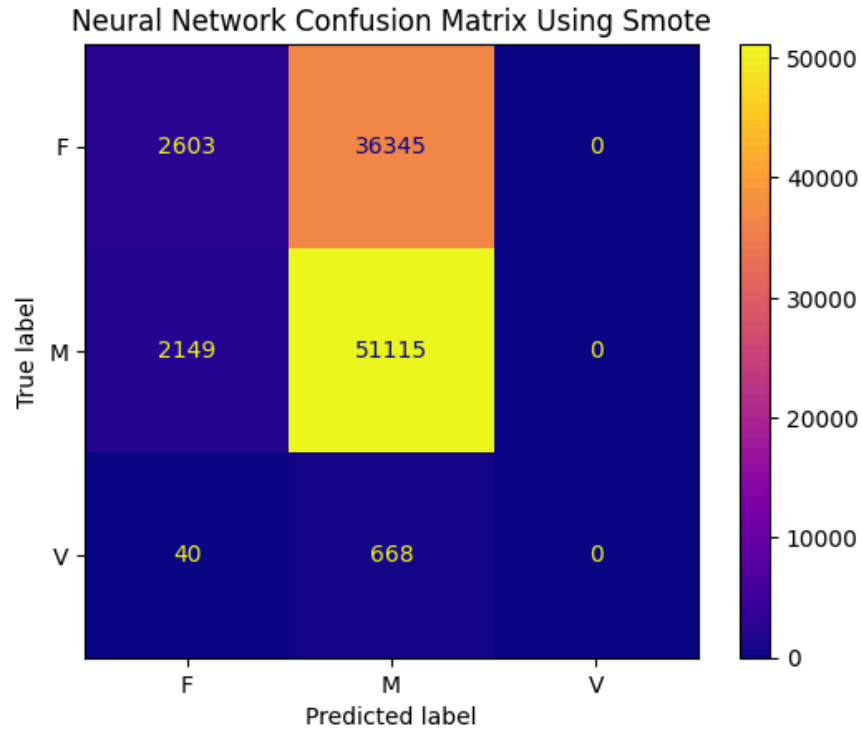
Neural Networks Visualizations & Results

Visualizations/Summary Statistics

	Accuracy	Balanced Accuracy
Entire Dataset	0.5754818216937803	0.25818388994284475
W/ SMOTE	0.5764476112804223	0.25662162498754804



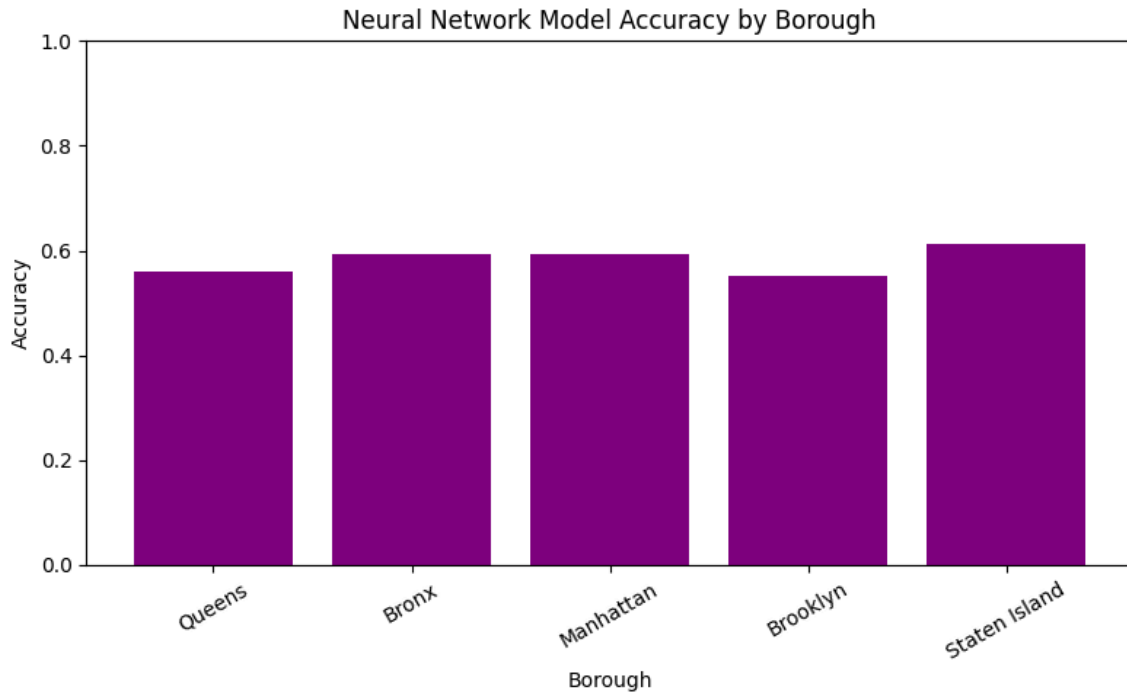
Higher values in the confusion matrix show a strong performance in classifying data for a specific label. Misdemeanors overtake the Neural Network’s predictions.



Higher values in the confusion matrix show a strong performance in classifying data for a specific label. Misdemeanors overtake the Neural Network's predictions using SMOTE.

Accuracy Without SMOTE separated By Borough:

Borough	Accuracy	Balanced Accuracy
Queens	0.5611089030206677	0.25676933417960496
Bronx	0.5931733884219916	0.2576604201274496
Manhattan	0.5929360425473885	0.259151340032283
Brooklyn	0.5512157167866174	0.2588178483462337
Staten Island	0.6130145427655903	0.25

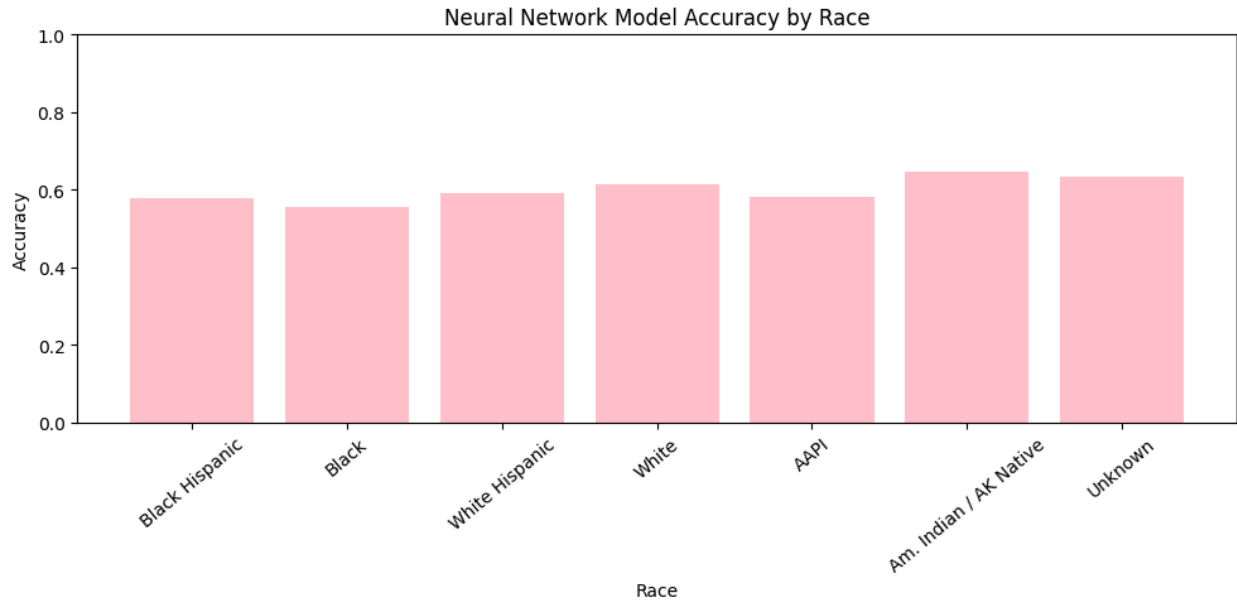


This bar chart depicts the neural network's accuracy grouped by borough. Staten Island, Bronx and Manhattan have the highest accuracies, but all borough accuracies range from 55-61%.

Accuracy Without SMOTE separated By Race:

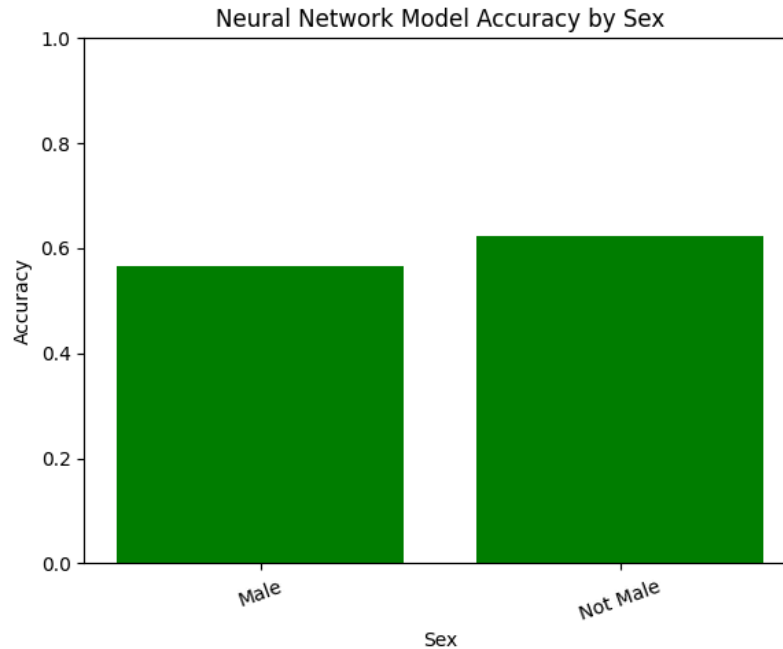
Race	Accuracy	Balanced Accuracy
Black & Hispanic	0.5782415918676328	0.25532778421997365
White	0.6123488224061108	0.25
AAPI	0.5818982013011864	0.2547369167275406
American Indian or Alaskan Native	0.6472491909385113	0.25117718446601944
Race unknown	0.6350148367952523	0.25
Black	0.5557179727291415	0.26215075488251677

White & Hispanic	0.5916486563634139	0.25
------------------	--------------------	------



Accuracy Without SMOTE separated By Sex:

Sex	Accuracy	Balanced Accuracy
Male	0.564862480744671	0.2589304111296468
Not Male	0.6222802436901653	0.25



This figure shows the neural network's accuracy grouped by sex. Not Male (Female) has a higher accuracy than the Male group.

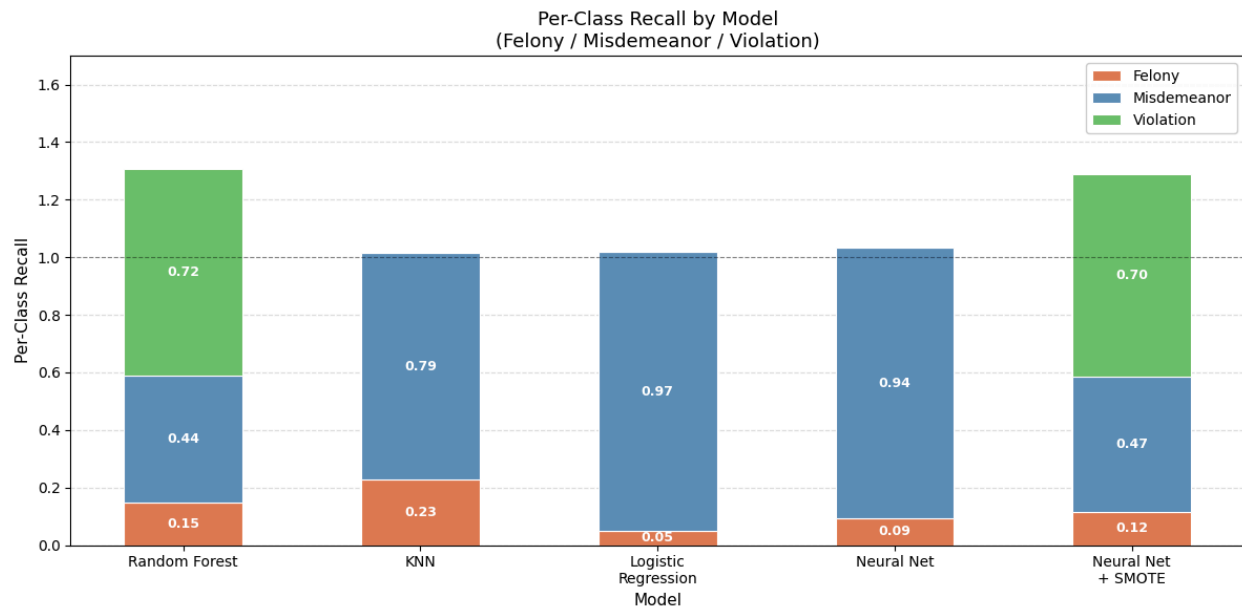
Further Discussion of Results

Overall, we saw that accuracy in models remained less than 60%, with most models having a general accuracy in the range of 55-58%. Balanced accuracy was much lower, ranging between 25%- 50%, with Random Forest Classifier having the highest balanced accuracy when tuned. This shows that the models are not accurate and are prone to predict misdemeanors, which are the majority class of offenses, since the balanced accuracy is low compared to the overall accuracy. For Random Forest Classifier, the baseline accuracy (57.6%) dropped after applying class balancing/tuning (25.2%), while balanced accuracy increased, indicating that class dominance of misdemeanors greatly affected our baseline accuracy. Similar trends followed for the rest of the models. Through analyzing the Logistic Regression model by subgroups of race, borough, and sex, we noticed that overall accuracy was highest in Brooklyn (54%) and Bronx (63%), but balanced accuracy was low between all subgroups, which remained consistent in other models.

Our models perform poorly across all demographic groups, and our overall accuracy score is attributable to misdemeanor class dominance, which our models tended to classify most data points as. Between all races and subgroups, all of our models perform poorly when it comes to balanced accuracy, even if they receive a high overall accuracy. As a result of this performance, we conclude that our features lack enough power to predict level of offense. It is possible that demographic disparities in charging exist but are unable to be explored by this distributional problem. Further, any variation in level of offense between boroughs is likely not attributable to any trends from demographic features, but result from data imbalance.

Before we arrived at our results, we experimented with different methods hoping to answer questions about societal trends in NYC based on policy change and year. We ended up not pursuing this research question due to limited time as well as complexity. We also wanted to predict future trends in criminal

arrests made by the NYPD across all boroughs, but we were unable to do so because of our model performances. Along with our research questions, we initially implemented KNN, Random Forest Classification, and Logistic Regression. However, when we calculated accuracies below 60% for all models, we decided to train a neural network to see if it could handle data and features that were more complex. We also calculated balanced accuracies for all models after noticing the discrepancy between level of offenses in the dataset, with most offenses being misdemeanors and only 1% being violations.



This stacked bar chart shows per-class recall for felony, misdemeanor, and violation across five models. Felony recall is consistently weak across all models (0.05–0.23), while misdemeanor recall dominates in Logistic Regression and Neural Net, reflecting the dataset's class imbalance. Only Random Forest and Neural Net + SMOTE capture meaningful violation recall (~0.70+), though SMOTE trades off misdemeanor performance to achieve this. No single model performs strongly across all three classes simultaneously.

Discussion & Conclusion

Our result suggests that the basic demographic features captured in NYPD arrest records are not sufficient on their own to predict how seriously a charge is classified. It is worth stating that this does not mean bias is absent from the arrest system, only that our model, with our features, is not adequate in answering the questions we intended to explore. Offense level at arrest is also not a fixed quantity, charges are routinely modified or dropped after arraignment, meaning our target variable itself may contain a significant amount of noise.

Our results connect to a broader conversation about using algorithms in criminal justice. Chouldechova (2017) showed that tools like COMPAS, which are used to predict recidivism risk, face a fundamental problem: when different racial groups have different base rates of reoffending, it is impossible for a model to be equally fair to all groups at the same time. COMPAS ends up with false positive rates for Black defendants nearly twice as high as for white defendants, not because the model is poorly built, but because of this tradeoff. Our project is not a recidivism tool, but it reinforces an important principle:

before any predictive model is used in a criminal justice context, rigorous fairness analysis across demographic subgroups is essential. Our results suggest that demographic information alone is a weak and potentially misleading basis for predicting offense level in this domain.

Reflection

Given another chance to approach this project, there are a variety of things that could be done differently. Firstly, we could incorporate precinct-level data and offense-type categories as features, since precinct-level enforcement cultures and specific offense types (e.g., drug vs. assault) can likely explain more variance in offense level than broad demographic categories. Another approach could be connecting arrest records to case outcomes, such as whether charges were reduced, dismissed, or resulted in conviction, would also allow researchers to trace disparities beyond the initial point of arrest. Another potential development could be situating our findings within specific policy periods (pre- and post-Stop-and-Frisk reform, pandemic-era policing changes) which could reveal patterns invisible in our selected two-year snapshot.

References

- Badr, Youakim, and Rahul Sharma. “Data Transparency and Fairness Analysis of the NYPD Stop-And-Frisk Program.” *Journal of Data and Information Quality*, vol. 14, no. 2, 30 June 2022, pp. 1–14, <https://doi.org/10.1145/3460533>.
- Chouldechova, A. (2017). Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2), 153–163. <https://doi.org/10.1089/big.2016.0047>
- Kumar, Jisha Sheela, et al. “View of Predictive Analytics in Law Enforcement: Unveiling Patterns in NYPD Crime through Machine Learning and Data Mining.” *Rebichte.org*, 2026, rebichte.org/index.php/rebichte/article/view/188/218. Accessed 27 Mar. 2026.